

Suggested Topics for FYP 2026-2027

MA Pui Kit Jack (Lecturer)
jackma@ie.cuhk.edu.hk

- Projects in Cryptography (cf. Prof. Sherman Chow):
 - constructing protocols to protect data from unauthorized read/write
 - math, computer science, algorithms, creative ideas, sense in "privacy"...
- Projects:
 - Forum/Web App Powered by **Anonymous Credential**
 - may start from **WebAuthn** (passwordless auth.) & extends to further protect the ID
 - why? get benefit from anonymity while preserving **accountability**; supported by blockchain e.g., <https://eprint.iacr.org/2021/1577>
 - **AI-powered** assistant app (e.g., prompt-engineering + system design)
 - focus on a specific topic, design a meaning project (or even an app that generate apps)
 - Example: birding app with location recommendation; read from social media (learning the trend) + weather + ...; detect bird direction from phone microphone; recognize bird from sound or photos (well, [this app](#) exists already); and features that could enrich the user experience and promoting data management
 - Self-proposed topics
 - related to **cryptography** (application of FHE, PSI, ZKP, ...), **blockchains** (may be), writing apps or AI-powered stuffs, ...

- For cryptography topics, expect to have **(strong) interest**
 - paper reading, coding & prototyping, math background
 - perhaps also a background (IERG4130, IERG4150, ...)
- For other topics, no general requirements (put efforts into FYP)
- Biweekly meeting
 - reporting progress and solving problems/brainstorming